

PRAKHAR MISHRA

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OBJECTIVE

Passionate and engineering-focused Electrical Engineering student seeking to contribute to the core technical team at AWS Student Builder Group HBTU. Eager to leverage strong computational mathematics logic, full-stack integration expertise, and hardware-automation skills to build community-driven tech architectures, organize technical hackathons, and manage student tech workflows.

TECHNICAL SKILLS

LANGUAGES : Python, Java, SQL, Vanilla JavaScript (ES6+), HTML5, CSS3, C

BACKEND & FRAMEWORKS : Spring Boot, REST API concepts, Web APIs, DOM Manipulation

HARDWARE & IOT : Arduino Uno, ESP32 microcontrollers, Serial communication(pyserial)

LIBRARIES & AUTOMATION : NumPy, Pandas, Pygame(UI Rendering), PyAutoGUI

TOOLS & HOSTING : Git, GitHub, Netlify, VS Code, IntelliJ IDEA, PyCharm, Arduino IDE, Tinkercad

PROJECTS

IOT NETWORK-CONTROLLED HARDWARE SWITCH | SPRING BOOT, PYSERIAL, ARDUINO :

- Developed a cross-platform IoT system utilizing Spring Boot to expose server-side logic controlling physical hardware switches.
- Programmed a custom Python script with pyserial for high-speed asynchronous data stream communication with an Arduino.
- Designed a responsive UI allowing any device on the local network to dynamically manipulate physical logic hardware via unified button interface. [Arduino-Web-Control](#)

MATRIX CALCULATOR PRO (COMPUTATIONAL MATH APP) | VANILLA JAVASCRIPT, CSS3, HTML5 :

- Built a web-based calculator performing advanced linear algebra algorithms, including matrix multiplication, determinants, and matrix inversion from scratch.
- Implemented recursive patterns using multi-dimensional JS arrays to compute determinants via minors and sub-matrices.
- Packaged modules enabling data outputs to be copied or downloaded instantly as formatted .json and text files. Hosted via Netlify. [Matrix-Calculator](#)

ULTRASONIC RADAR TRACKING TOOL | PYSERIAL, PYGAME, ARDUINO UNO :

- Formulated a real-time radar system using an ultrasonic sensor rotated via a 9g micro-servo motor and used pyserial to extract servo position and objects within a specified threshold.
- Programmed a graphics interface in Python with Pygame to visualise the received data and dynamically update the screen. [Radar](#)

ACHIEVEMENTS :

DATA STRUCTURES & ALGORITHMS :

Solved 50+ LeetCode problems, demonstrating strong proficiency in array manipulation, recursion, and optimisation. [LeetCode](#)

DEPLOYMENT :

Designed and launched a personal portfolio website to host project source files, deployed via Netlify. [Portfolio](#)